

Proof versus Pattern Student Handout

TOK Cognitive Science Activity Bank • Mathematics • 30-40 min

Category	Mathematics	Time	30-40 min
Difficulty	Medium	ID	proof-versus-pattern

Big idea: Examples are not proof.

Student-Facing Prompt

Inspect Proof versus Pattern Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Proof versus Pattern Stimulus A	Student-facing stimulus 1: A number pattern such as $1 + 2 + \dots + n = n(n+1)/2$ that invites both pattern-spotting and proof. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Proof versus Pattern Stimulus B	Student-facing stimulus 2: A visual pattern students extend before testing whether it proves anything. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Proof versus Pattern Reveal / Twist	Student-facing stimulus 3: A comparison between inductive confidence and deductive proof. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Examples are not proof.

Actual Lesson Questions

#	Question for students
1	After inspecting Proof versus Pattern Stimulus A, what is your first knowledge claim?
2	Which exact detail from Proof versus Pattern Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	Why is proof valued more than observation in mathematics?

Ready-to-Use Examples

- A number pattern such as $1 + 2 + \dots + n = n(n+1)/2$ that invites both pattern-spotting and proof.
- A visual pattern students extend before testing whether it proves anything.
- A comparison between inductive confidence and deductive proof.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Activity Steps

1. Students calculate $n^2 + n + 41$ for $n = 0, 1, 2, \dots$ and record whether each output is prime.
2. They predict whether the expression always gives primes.
3. Reveal a counterexample, such as $n = 40$.
4. Students write the difference between many examples and proof.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: Why is proof valued more than observation in mathematics?
- Can something be obvious but still require proof?
- Is mathematics discovered or invented?

Knowledge Questions

- Why is proof valued more than observation in mathematics?
- Can something be obvious but still require proof?
- Is mathematics discovered or invented?

Transfer Example

For example, run this activity with a number pattern such as $1 + 2 + \dots + n = n(n+1)/2$ that invites both pattern-spotting and proof. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...